Hepatic Vessel and Tumor Segmentation Using 3D nnU-Net

Ankith Janumpally
Department of Biomedical Engineering,
Case Western Reserve University

ABSTRACT

Automatic segmentation of hepatic vessels and tumors from computed tomography (CT) can support treatment planning and image-guided procedures by providing patient-specific maps of tumor location and surrounding vasculature. However, hepatic vessel segmentation remains challenging because vessels form thin, branching structures in which small prediction errors can interrupt vessel continuity. This study evaluated a baseline 3D nnU-Net for simultaneous hepatic vessel and tumor segmentation using the Medical Segmentation Decathlon Task08 Hepatic Vessel dataset. Of 303 labeled CT volumes, 242 were used for model development and 61 were reserved as an independent holdout cohort. The baseline model was trained using the 3D full-resolution nnU-Net configuration with Dice and cross-entropy loss. A second model incorporated a vessel-specific topology-aware clDice term with a weight of 0.3 while maintaining the same architecture, preprocessing, data split, and training settings. On the holdout cohort, the baseline achieved Dice scores of 0.6351 for vessels and 0.6678 for tumors, with a mean foreground Dice of 0.6514. Adding clDice increased these values to 0.6373, 0.6698, and 0.6536, respectively. These results demonstrate a modest improvement with topology-aware supervision, although further tuning and topology-specific evaluation are needed to determine whether vessel connectivity is meaningfully improved.

1. INTRODUCTION

Accurate identification of hepatic vessels and tumors is important for understanding patient-specific liver anatomy and can support applications such as surgical planning, treatment planning, and image-guided intervention. Because tumors may lie near or involve major vascular structures, delineating both the lesion and surrounding vasculature can help characterize their spatial relationship before treatment. Automated segmentation provides a way to generate these anatomical maps directly from medical images and may reduce the time and variability associated with manual delineation.[1]

Hepatic vessel segmentation from computed tomography (CT) is technically challenging because the vascular network consists of thin, branching structures that occupy a relatively small portion of the image volume. Small distal branches may span only a few voxels, making them particularly susceptible to missed detections or discontinuities in the predicted segmentation. In addition, vessel size, contrast, and branching patterns vary across patients. Recent deep learning approaches have demonstrated the potential for automated hepatic vessel segmentation, but maintaining accurate delineation of small branches and vascular continuity remains an important challenge.[2]

Deep learning methods, particularly convolutional neural networks, are widely used for medical image segmentation because they can learn complex spatial and intensity features directly from annotated imaging data. In this project, a 3D nnU-Net was used as the baseline model for simultaneous segmentation of hepatic vessels and tumors. Conventional overlap-based objectives reward agreement between predicted and ground-truth voxels but do not explicitly account for the connectivity of tubular structures.

Therefore, this study compared a baseline 3D nnU-Net with a modified model incorporating a vessel-specific clDice loss. It was hypothesized that adding topology-aware clDice supervision to the standard nnU-Net loss would improve hepatic vessel segmentation performance on an independent holdout cohort while maintaining tumor segmentation performance.

2. MATERIALS AND METHODS

2.1 Dataset and Data Split

The Medical Segmentation Decathlon Task08 Hepatic Vessel dataset was used for this study.[3] The dataset contains 303 labeled abdominal CT volumes with voxel-level annotations for hepatic vessels and tumors. Labels were encoded as background, vessel, and tumor classes. To create an independent evaluation cohort, the 303 labeled cases were divided into 242 development cases and 61 holdout cases using a fixed random seed. The 61 holdout cases were excluded from model training and validation. Within the development cohort, nnU-Net fold 0 contained 193 training cases and 49 validation cases.

2.2 Preprocessing

Preprocessing was performed using the nnU-Net v2 pipeline.[4] CT intensities were normalized, and the volumes were resampled to a common target spacing of approximately 1.5 × 0.80 × 0.80 mm. The data were then prepared for three-dimensional patch-based training using patches of 64 × 192 × 192 voxels. These preprocessing steps standardized image properties across cases while allowing large CT volumes to be processed within available GPU memory. The same preprocessing pipeline was used for both the baseline and modified models.

2.3 Baseline nnU-Net

The baseline model used the 3D full-resolution nnU-Net configuration with a PlainConvUNet architecture.[4] Training was performed on fold 0 for 1000 epochs with a batch size of 2. The network contained six resolution stages with feature map sizes of 32, 64, 128, 256, 320, and 320 channels and approximately 88.2 million trainable parameters. The baseline objective combined Dice loss and cross-entropy loss for multiclass segmentation of hepatic vessels and tumors.

Table 1. Baseline nnU-Net configuration and training hyperparameters

Parameter	Value
Configuration	3D full resolution
Architecture	PlainConvUNet
Training fold	Fold 0
Training cases	193
Validation cases	49
Epochs	1000
Patch size	64 × 192 × 192
Batch size	2
Loss function	Dice + cross entropy
Target spacing	1.5 × 0.80 × 0.80 mm
Trainable parameters	~88.2 million

2.4 clDice Modification

To extend the baseline model, a vessel-specific topology-aware clDice term was added to the original Dice and cross-entropy objective. clDice was developed to encourage preservation of topology in tubular structures by comparing the centerline structure of the prediction and ground truth.[5] A clDice weight of 0.3 was used, producing a combined loss consisting of the original nnU-Net loss plus the weighted vessel clDice term.

The clDice component was applied only to the vessel class because vessel connectivity was the primary structural feature of interest. To reduce GPU memory requirements, vessel probabilities and ground-truth masks were downsampled by a factor of 2 before calculation of the topology term, and five soft-skeletonization iterations were used. All other training settings, including the architecture, data split, preprocessing, patch size, batch size, and number of epochs, were kept unchanged to provide a controlled comparison with the baseline.

2.5 Evaluation

Model performance was evaluated on the independent 61-case holdout cohort using the Dice similarity coefficient. Dice scores were calculated separately for the vessel and tumor classes, and the mean foreground Dice was also reported to summarize overall segmentation performance.

In addition to aggregate performance, per-case Dice scores were calculated for all holdout cases to examine variability in model performance. Qualitative comparisons were then generated for selected low-performing, high-performing, and clDice-improved cases. Each comparison included the original CT image, ground-truth segmentation, baseline nnU-Net prediction, and nnU-Net + clDice prediction.

3. RESULTS

3.1 Quantitative Performance

Both models were evaluated on the same 61-case independent holdout cohort. The baseline nnU-Net achieved a vessel Dice of 0.6351 and a tumor Dice of 0.6678, resulting in a mean foreground Dice of 0.6514. After adding the vessel-specific clDice term, vessel Dice increased to 0.6373 and tumor Dice increased to 0.6698, producing a mean foreground Dice of 0.6536. Overall, clDice improved mean holdout Dice by 0.0021, with small improvements observed for both foreground classes.

Table 2. Baseline and clDice performance on the independent holdout cohort

Metric	Baseline nnU-Net	nnU-Net+clDice
Vessel Dice	0.6351	0.6373
Tumor Dice	0.6678	0.6698
Mean foreground Dice	0.6514	0.6536

Interestingly, the internal fold-0 validation Dice was slightly higher for the baseline model than for the clDice model, with values of 0.6782 and 0.6740, respectively. However, the clDice model produced slightly better performance on the independent holdout cohort. This difference emphasizes the importance of evaluating both models on data that were not used during model development.

3.2 Qualitative Results

Qualitative evaluation demonstrated substantial variation in segmentation performance across the independent holdout cohort. A low-performing example was hepaticvessel_075, for which the baseline model achieved a vessel Dice of 0.0206 and a tumor Dice

of 0.0000, resulting in a mean Dice of 0.0103. As shown in Fig. 1, the ground-truth segmentation contains both a distinct tumor region and an adjacent vessel structure, while the baseline and clDice models produced fragmented predictions with minimal overlap with the annotated regions. This case illustrates a major segmentation failure and helps explain why the overall holdout Dice remained moderate despite much stronger performance in other cases.

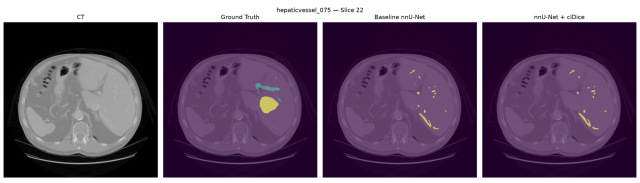


Figure 1. Poor-performing holdout case with minimal overlap between predictions and ground truth.

In contrast, hepaticvessel_110 represented a high-performing case. The baseline model achieved a vessel Dice of 0.8021 and a tumor Dice of 0.9345, corresponding to a mean Dice of 0.8683. As shown in Fig. 2, both the baseline and clDice models closely reproduced the primary tumor region and captured the major vessel regions visible in the selected axial slice. This example demonstrates that the same segmentation framework was capable of producing highly accurate predictions in favorable cases.

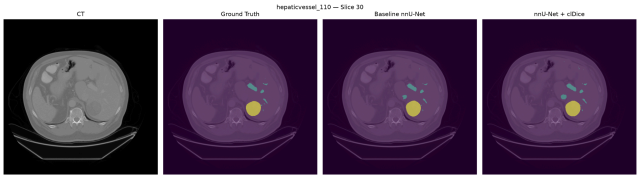


Figure 2. High-performing holdout case with close agreement between predictions and ground truth.

The largest improvement with clDice was observed for hepaticvessel_308. For this case, mean Dice increased from 0.4065 with the baseline model to 0.5046 with the modified model, corresponding to an absolute improvement of 0.0982. As shown in Fig. 3, differences between the baseline and clDice predictions on the selected axial slice are relatively subtle, particularly for the small vessel regions. Because Dice was calculated over the complete three-dimensional volume, the larger numerical improvement likely reflects differences distributed across multiple slices rather than a single large local correction. Therefore, this case demonstrates an improvement in segmentation performance with clDice, although the present qualitative analysis alone cannot determine whether vessel topology or connectivity was specifically improved.

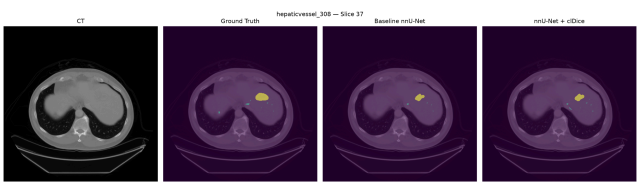


Figure 3. Holdout case with the largest improvement after adding clDice.

Overall, these examples illustrate the wide range of segmentation performance observed across the holdout cohort. The poor-performing case demonstrates the types of errors that reduce the overall Dice score, while the high-performing case shows that accurate vessel and tumor segmentation was achievable for some patients. The clDice-improved case further shows that the modified loss could provide meaningful case-specific improvements even though the average improvement across the full holdout cohort was small.

4. DISCUSSION

The results partially supported the original hypothesis that adding vessel-specific clDice supervision would improve segmentation performance compared with the baseline nnU-Net. On the independent holdout cohort, the clDice model increased vessel Dice from 0.6351 to 0.6373 and mean foreground Dice from 0.6514 to 0.6536. Tumor Dice also increased slightly from 0.6678 to 0.6698, despite clDice being applied only to the vessel class. However, these improvements were small, and the baseline model actually produced a slightly higher internal validation Dice than the clDice model. This suggests that the topology-aware modification did not produce a large or consistent improvement across all evaluation settings.

The modest change in Dice is not necessarily inconsistent with the intended purpose of clDice. Standard Dice primarily measures voxel overlap, whereas clDice was developed to preserve the topology of tubular structures by emphasizing agreement between predicted and ground-truth centerlines.[5] As a result, a model could potentially improve vessel continuity without producing a large change in conventional Dice. This distinction is especially important for hepatic vessels, where segmentation errors may occur as small gaps or missing branches rather than large errors in overall vessel volume. Prior work on hepatic vessel segmentation has similarly identified small vessels and complex branching anatomy as persistent challenges for automated methods.[2,7] In addition, Dice alone does not fully describe properties such as connectivity, boundary accuracy, or topology, which limits the conclusions that can be drawn from overlap-based evaluation alone.[6]

Several experimental limitations may also explain the limited improvement. Only a single clDice weight of 0.3 was evaluated, so the balance between conventional overlap loss and topology-aware supervision was not optimized. The clDice component was also computed at half resolution to remain within GPU memory constraints, which may have reduced its sensitivity to small vessel branches. In addition, only one nnU-Net fold and one modified loss configuration were tested rather than a broader hyperparameter search. Although nnU-Net provides a strong automatically configured baseline for biomedical segmentation, its performance can still depend on dataset-specific training and evaluation choices.[4]

Overall, the experiment demonstrates that a topology-aware loss can be incorporated into a standard 3D nnU-Net pipeline without sacrificing holdout performance, but the observed benefit was modest. Future work should evaluate multiple clDice weights, calculate topology-specific metrics such as clDice directly, and perform a systematic failure-case analysis across the full holdout cohort. Additional qualitative analysis of low-performing cases may also help determine whether the remaining errors are

primarily caused by missed small vessels, fragmented branches, difficult tumor boundaries, or limited image contrast.

5. CONCLUSION

This study evaluated a baseline 3D nnU-Net and a modified model incorporating vessel-specific clDice for simultaneous hepatic vessel and tumor segmentation from CT. The baseline achieved a mean foreground Dice of 0.6514 on the independent holdout cohort, while the clDice model modestly improved performance to 0.6536. Although the improvement was small, the modified model increased Dice for both vessel and tumor segmentation without reducing holdout performance.

These findings suggest that topology-aware supervision may provide some benefit for hepatic vessel segmentation, particularly for thin and branching vascular structures. However, the effect observed in this study was limited, and Dice alone cannot determine whether vessel connectivity was meaningfully improved. Further work should evaluate additional clDice weights, topology-specific metrics, and qualitative failure cases to better assess the value of topology-aware loss functions for hepatic vessel segmentation.

6. REFERENCES

1. Oliveira DAB, Feitosa RQ, Correia MM. Segmentation of liver, its vessels and lesions from CT images for surgical planning. BioMed Eng OnLine. 2011;10:30. doi:10.1186/1475-925X-10-30.
2. Sarkar S, Rahmani M, Farnia P, Ahmadian A, Mozayani N. Automated multiclass segmentation of liver vessel structures in CT images using deep learning approaches: a liver surgery pre-planning tool. Phys Eng Sci Med. 2025;48(3):1463-1472. doi:10.1007/s13246-025-01581-7.
3. Antonelli M, Reinke A, Bakas S, et al. The Medical Segmentation Decathlon. Nat Commun. 2022;13:4128. doi:10.1038/s41467-022-30695-9.
4. Isensee F, Jaeger PF, Kohl SAA, Petersen J, Maier-Hein KH. nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation. Nat Methods. 2021;18(2):203-211. doi:10.1038/s41592-020-01008-z.
5. Shit S, Paetzold JC, Sekuboyina A, et al. clDice—a novel topology-preserving loss function for tubular structure segmentation. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. 2021:16560-16569. doi:10.1109/CVPR46437.2021.01629.
6. Müller D, Soto-Rey I, Kramer F. Towards a guideline for evaluation metrics in medical image segmentation. BMC Res Notes. 2022;15(1):210. doi:10.1186/s13104-022-06096-y.
7. Kuang H, Yang Z, Zhang X, Tan J, Wang X, Zhang L. Hepatic vein and arterial vessel segmentation in liver tumor patients. Comput Intell Neurosci. 2022;2022:2303733. doi:10.1155/2022/2303733.